

Module 1: Lesson 2

Characteristic Functions



Outline

- ▶ Characteristic functions.
- ▶ Deriving Characteristic functions: Black-Scholes.

Characteristic functions

As we have already seen, one important input in Fourier-type option pricing methods is the characteristic function (CF) of the underlying stock price process.

The CF will come in handy because, for complex methods, directly observing the CDF is impossible and/or too computationally costly. In this Lesson, we will go over the mathematical foundations of CF, and see an step-by-step example on how to derive it for a simple (Black-Scholes type) stock price process.

The **characteristic function**, $\varphi_X(u)$, of a real valued random variable X depicts the **distribution function uniquely**. It is defined as:

$$\varphi_X(u) = \mathbb{E}[e^{iuX}] , \text{ with } i = \sqrt{-1}$$

Conveniently, we can rewrite this as:

$$\varphi_X(u) = \int_{-\infty}^{\infty} e^{iuX} dF(x) = \int_{-\infty}^{\infty} e^{iuX} f(x) dx$$

where $F(x)$ and $f(x)$ denote the CDF and PDF of X , respectively.

Deriving characteristic functions

Next, we will go over the process of deriving an specific characteristic function from scratch. We will do that for the **Black-Scholes** model assumptions. However, the process is common to other models. Before doing that, let's define a **Lemma** that will help us later on in the derivation.

► **Lemma.** Define $y = \hat{\mu} + iu\hat{\sigma}^2$, then:

$$ius_T - \frac{(s_T - \hat{\mu})^2}{2\hat{\sigma}^2} = \frac{-(s_T - y)^2 + 2\hat{\mu}\hat{\sigma}^2 iu - u^2\hat{\sigma}^4}{2\hat{\sigma}^2}$$

Proof.

$$\begin{aligned} ius_T - \frac{(s_T - \hat{\mu})^2}{2\hat{\sigma}^2} &= \frac{2\hat{\sigma}^2 ius_T - (s_T^2 - 2s_T\hat{\mu} + \hat{\mu}^2)}{2\hat{\sigma}^2} = \frac{-s_T^2 + 2s_T\hat{\mu} + 2s_T iu\hat{\sigma}^2 - \hat{\mu}^2}{2\hat{\sigma}^2} = \\ &= \frac{-(s_T - y)^2 + 2\hat{\mu}\hat{\sigma}^2 iu - u^2\hat{\sigma}^4}{2\hat{\sigma}^2} \end{aligned}$$

Deriving Black-Scholes CF

Now, we will work on deriving the CF of a log-normal process, as it is the case with stock prices in BS.

Let's start by working with a normally-distributed variable $s = \ln S$, with $\mathbb{Q}(s_T)$ its risk-neutral PDF:

$$\begin{aligned}\varphi_{s_T}(u) &= \mathbb{E}[e^{ius_T}] = \int_{-\infty}^{\infty} e^{ius_T} \mathbb{Q}(s_T) ds_T = \int_{-\infty}^{\infty} e^{ius_T} \frac{1}{\sigma\sqrt{2\pi T}} e^{-\frac{\{s_T - [s_0 + (\mu - \frac{\sigma^2}{2})T]\}^2}{2\sigma^2 T}} ds_T = \\ &= \frac{1}{\sigma\sqrt{2\pi T}} \int_{-\infty}^{\infty} e^{ius_T - \frac{\{s_T - [s_0 + (\mu - \frac{\sigma^2}{2})T]\}^2}{2\sigma^2 T}} ds_T\end{aligned}$$

Let $\hat{\mu} = [s_0 + (\mu - \frac{\sigma^2}{2})T]$ and $\hat{\sigma} = \sigma\sqrt{T}$, then:

$$\varphi_{s_T}(u) = \frac{1}{\sigma\sqrt{2\pi T}} \int_{-\infty}^{\infty} e^{ius_T - \frac{\{s_T - [s_0 + (\mu - \frac{\sigma^2}{2})T]\}^2}{2\sigma^2 T}} ds_T = \frac{1}{\hat{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ius_T - \frac{(s_T - \hat{\mu})^2}{2\hat{\sigma}^2}} ds_T$$

Deriving Black-Scholes CF

Now, the **Lemma** defined before comes in handy. Applying it:

$$\begin{aligned}\varphi_{s_T}(u) &= \frac{1}{\hat{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ius_T - \frac{(s_T - \hat{\mu})^2}{2\hat{\sigma}^2}} ds_T = \frac{1}{\hat{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{-(s_T - y)^2 + 2\hat{\mu}\hat{\sigma}^2 iu - u^2 \hat{\sigma}^4}{2\hat{\sigma}^2}} ds_T = \\ &= e^{\frac{2\hat{\mu}\hat{\sigma}^2 iu - u^2 \hat{\sigma}^4}{2\hat{\sigma}^2}} \frac{1}{\hat{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{-(s_T - y)^2}{2\hat{\sigma}^2}} ds_T = e^{\hat{\mu}iu - \frac{1}{2}u^2 \hat{\sigma}^2} \frac{1}{\hat{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{s_T - y}{\sqrt{2\hat{\sigma}^2}}\right)^2} ds_T\end{aligned}$$

Now, consider $z = \frac{s_T - y}{\sqrt{2\hat{\sigma}^2}}$, and closed upper and lower bounds of s_T , $[-\gamma, \gamma]$,

$$\begin{aligned}z &= \frac{s_T - (\hat{\mu} + iu\hat{\sigma}^2)}{\sqrt{2\hat{\sigma}^2}} = \frac{s_T - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}} - \frac{iu\hat{\sigma}}{\sqrt{2}} \\ \varphi_{s_T}(u) &= e^{\hat{\mu}iu - \frac{1}{2}u^2 \hat{\sigma}^2} \frac{1}{\hat{\sigma}\sqrt{2\pi}} \int_{\frac{-\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}} - \frac{iu\hat{\sigma}}{\sqrt{2}}}^{\frac{\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}} - \frac{iu\hat{\sigma}}{\sqrt{2}}} e^{-z^2 \sqrt{2\hat{\sigma}^2}} dz\end{aligned}$$

Deriving Black-Scholes CF

Cauchy's integral theorem. If a function $f(z)$ is analytic in a simply connected domain Ω , then for any simply closed contour C in Ω ,

$$\int_C f(z) dz = 0$$

Let a_i , for $i = 1, 2, 3, 4$ denote the corners of the rectangular contour C :

$$a_1 = \frac{\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}, \quad a_2 = \frac{-\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}, \quad a_3 = \frac{-\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}} - \frac{iu\hat{\sigma}}{\sqrt{2}}, \quad a_4 = \frac{\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}} - \frac{iu\hat{\sigma}}{\sqrt{2}}$$

By Cauchy's theorem we know that,

$$\oint_{C \in \Omega} \sqrt{2\hat{\sigma}^2} e^{-z^2} dz = 0$$

$$\Leftrightarrow \int_{a_1}^{a_2} \sqrt{2\hat{\sigma}^2} e^{-z^2} dz + \int_{a_2}^{a_3} \sqrt{2\hat{\sigma}^2} e^{-z^2} dz + \int_{a_3}^{a_4} \sqrt{2\hat{\sigma}^2} e^{-z^2} dz + \int_{a_4}^{a_1} \sqrt{2\hat{\sigma}^2} e^{-z^2} dz = 0$$

Deriving Black-Scholes CF

We can now solve the previous integrals separately and check their behavior when $\gamma \rightarrow \infty$. Let's start with those that do not contain any imaginary part in the bounds:

$$\lim_{\gamma \rightarrow \infty} \int_{a_1}^{a_2} \sqrt{2\hat{\sigma}^2} e^{-z^2} dz = - \lim_{\gamma \rightarrow \infty} \sqrt{2\hat{\sigma}^2} \int_{a_2}^{a_1} e^{-z^2} dz = -\sqrt{2\pi\hat{\sigma}^2}$$

Since we know that $\int_{-\infty}^{\infty} e^{-z^2} dz = \sqrt{\pi}$. Next, let's check those integrals with imaginary component in only one of the bounds, for example that between a_4 and a_1 :

$$\lim_{\gamma \rightarrow \infty} \int_{a_4}^{a_1} \sqrt{2\hat{\sigma}^2} e^{-z^2} dz = \lim_{\gamma \rightarrow \infty} \sqrt{2\hat{\sigma}^2} \int_{a_4}^{a_1} e^{-z^2} dz = 0$$

It is easy to check that these will be 0, since we know that, for example, $\int_{\infty-iz}^{\infty} e^{-z} dz = 0$. Finally, to fulfill Cauchy's theorem we know that:

$$\int_{a_3}^{a_4} \sqrt{2\hat{\sigma}^2} e^{-z^2} dz = \sqrt{2\pi\hat{\sigma}^2}$$

Deriving Black-Scholes CF

Plugging the previous result on the characteristic function expression we had from before:

$$\varphi_{s_T}(u) = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2} \frac{1}{\hat{\sigma}\sqrt{2\pi}} \int_{\frac{-\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}} - \frac{iu\hat{\sigma}}{\sqrt{2}}}^{\frac{\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}} - \frac{iu\hat{\sigma}}{\sqrt{2}}} e^{-z^2} \sqrt{2\hat{\sigma}^2} dz = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2} \frac{\sqrt{2\pi\hat{\sigma}^2}}{\hat{\sigma}\sqrt{2\pi}} = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2}$$

Replacing back $\hat{\mu} = \left[s_0 + \left(\mu - \frac{\sigma^2}{2} \right) T \right]$ and $\hat{\sigma} = \sigma\sqrt{T}$, yields **Black-Scholes CF**:

$$\varphi_{s_T}(u) = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2} = e^{\left(s_0 + \left(\mu - \frac{\sigma^2}{2} \right) T \right) iu - \frac{1}{2}u^2\sigma^2 T}$$

Summary of Lesson 2

In Lesson 2 we have looked at:

- ▶ What are characteristic functions and why are they helpful in Fourier pricing methods.
- ▶ Derive from scratch the characteristic function of a Black-Scholes type process.

⇒ **TO-DO NEXT:** This Lesson does not have an associated **Jupyter Notebook**. You can however check the extra readings and additional material provided to see how to work with more complex characteristic functions.

⇒ **In the Next Lesson** we will take a closer look at the math behind the **Discrete Fourier Transform (DFT)** and the **Fast Fourier Transform (FFT)**.